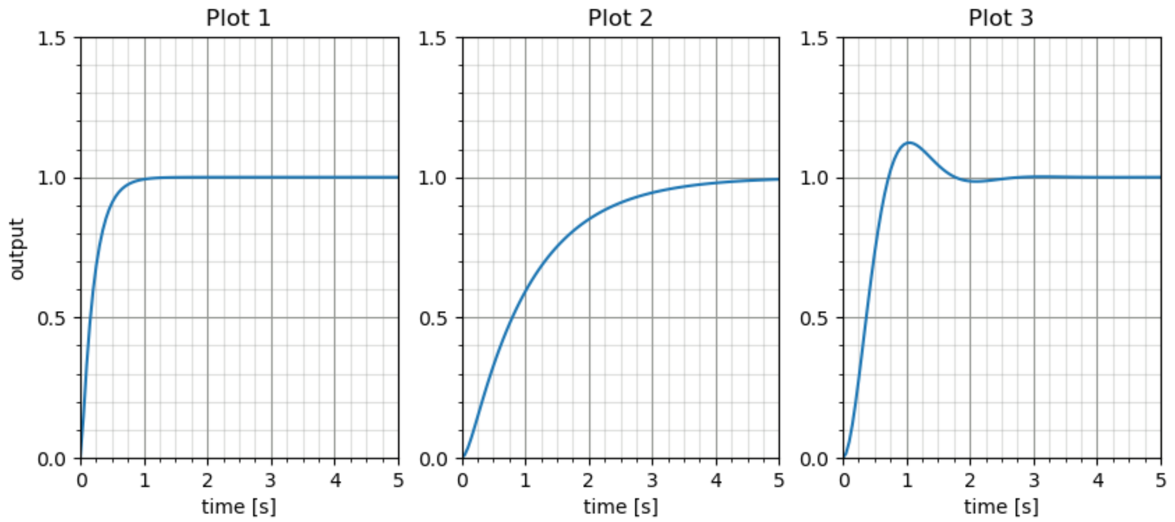
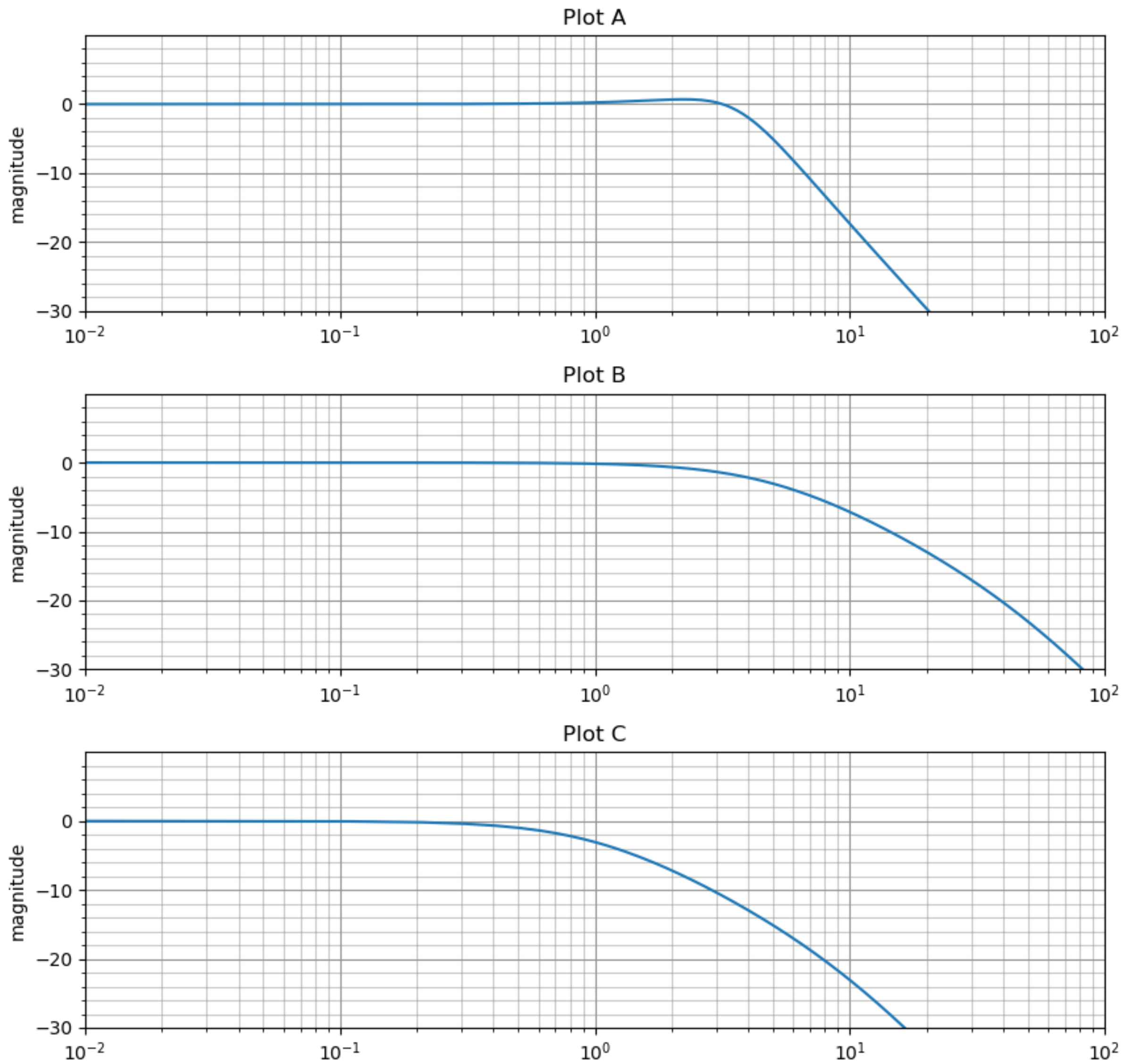
Tutorial 3: System Identification – Frequency Domain

Question 1: time and frequency responses

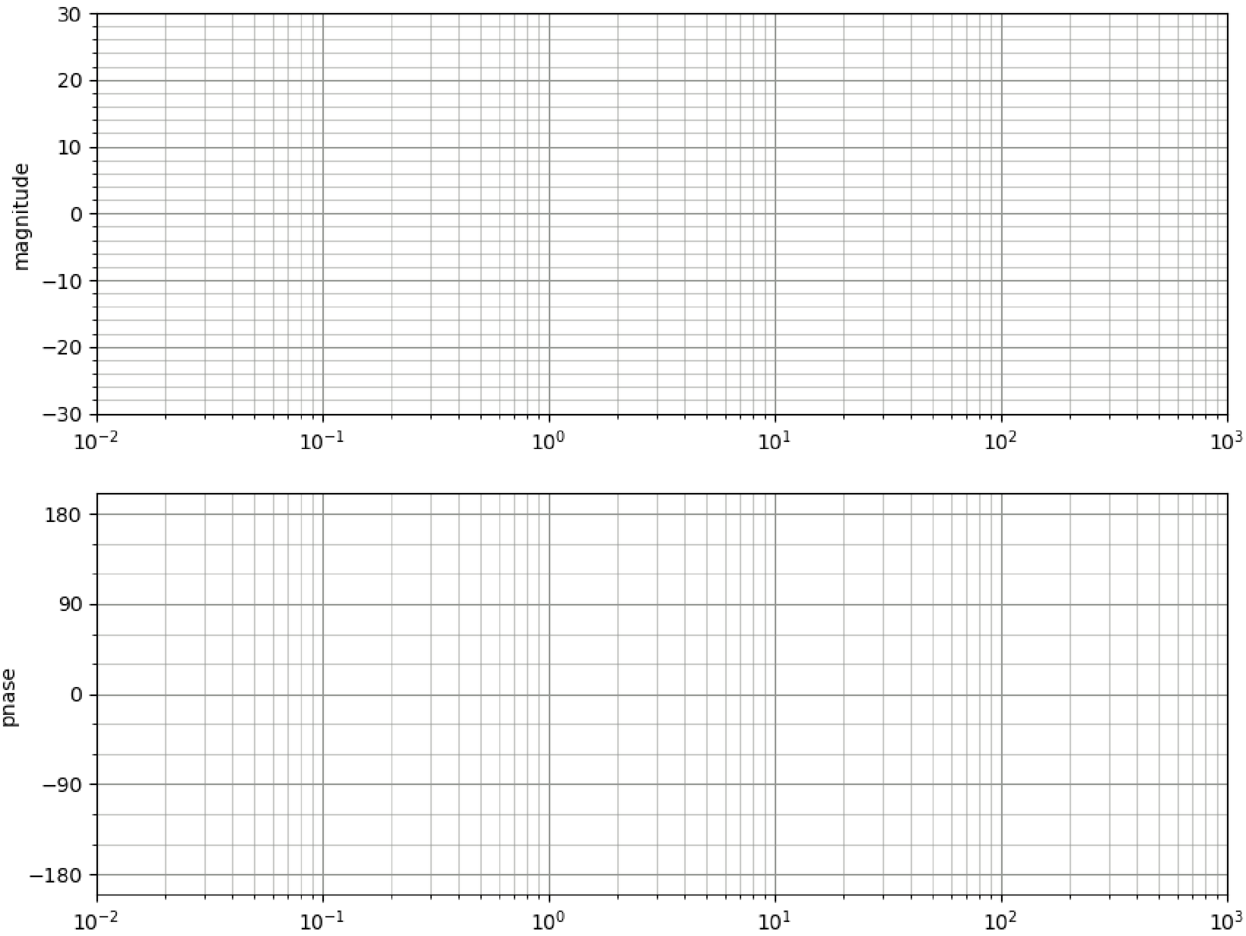
Each of the following plots 1-3 represents the unit step response for a second-order system, while the plots A-C give the magnitudes of their frequency responses. Match the corresponding plots, briefly justifying your selections.



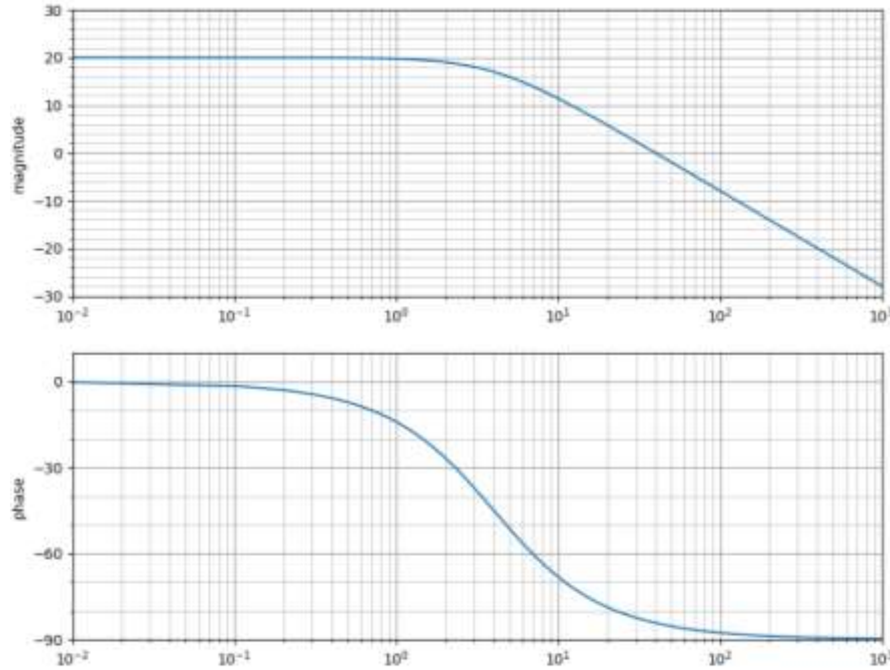


Question 2: first-order frequency responses

a) Give a rough* sketch of the frequency response of the system $G(s) = \frac{20}{s+10}$ on the Bode plot below. Check your work by selecting five frequencies spanning a sensible range and calculating the magnitude and phase of the frequency response for these values. Show the resulting points on your plot.
(rough* = asymptotic. We want to see the low and high frequency asymptotes, and the corner frequency)



b) Determine the transfer function of the first-order system with the following frequency response:

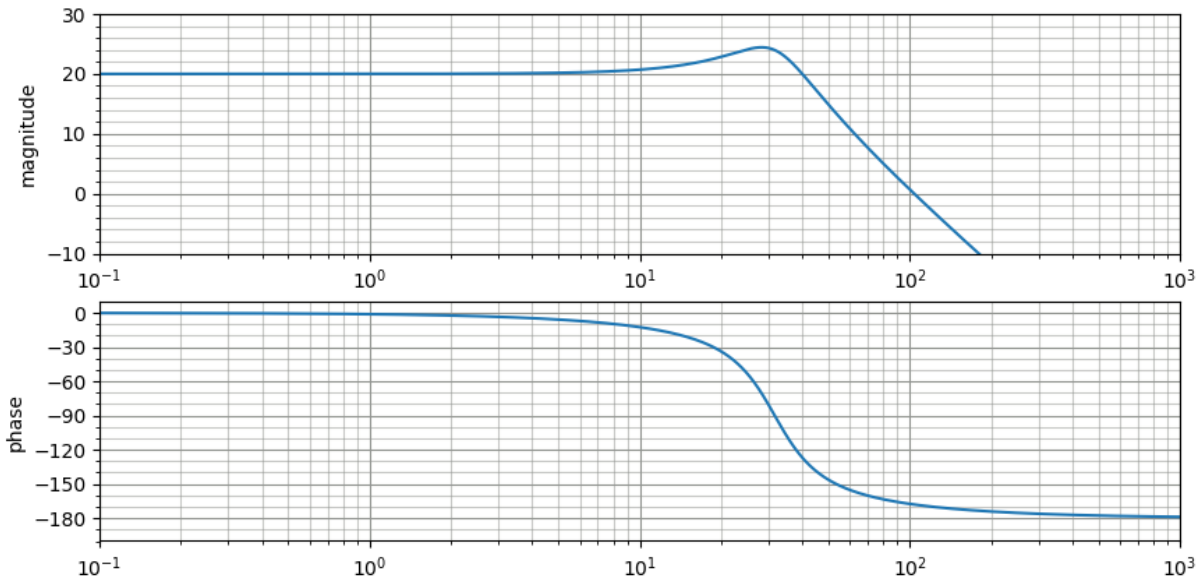


Question 3: second-order frequency responses

The Bode plots below were produced by second-order systems. Based on these responses,

i) Do you expect the transfer function to have two real poles, or a complex pole pair? Explain.
ii) Determine the transfer function of the system.

Plot A



Plot B